Unit: Unit 5: Linear Equations
Topic: Scatter Plot and The Best Fit Line

Name: ____________
Date: ________

Foundation (Jalapeno)

Q1. What is the purpose of a scatter plot?
(A) To show the relationship between two variables
(B) To display a single data point
(C) To create a line graph
(D) To make predictions
(E) To show the average of a dataset

Q2. Which type of correlation is shown by the points $(1,2),(2,4),(3,6)$?
(A) Positive
(B) Negative
(C) None
(D) Inverse
(E) Direct

Q3. What is the equation of the line of best fit for the points $(0,0),(1,2),(2,4)$?
(A) $y=x$
(B) $y=2x$
(C) $y=x+1$
(D) $y=2x-1$
(E) $y=x-2$

Q4. If a line of best fit has a positive slope, what can be said about the relationship between the variables?
(A) As one variable increases, the other decreases
(B) As one variable increases, the other increases
(C) There is no relationship between the variables
(D) The relationship is inversely proportional
(E) The relationship is directly proportional

Q5. What is the purpose of drawing a line of best fit?
(A) To show the average of a dataset
(B) To make predictions based on data trends
(C) To display a single data point
(D) To create a scatter plot
(E) To show the relationship between two variables

Q6. If a scatter plot shows a strong positive correlation, what can be said about the slope of the line of best fit?
(A) The slope is negative
(B) The slope is zero
(C) The slope is positive
(D) The slope is undefined
(E) The slope is inversely proportional

Q7. Which of the following is an example of a prediction based on a line of best fit?
(A) Using a line of best fit to find the average of a dataset
(B) Using a line of best fit to estimate a value outside the range of the data
(C) Using a line of best fit to display a single data point
(D) Using a line of best fit to show the relationship between two variables
(E) Using a line of best fit to create a scatter plot

Q8. If a line of best fit has an equation of $y=2x+1$, what is the predicted value of y when $x=3$?
(A) $y=5$
(B) $y=6$
(C) $y=7$
(D) $y=8$
(E) $y=9$

Open Ended Questions (FRQ)

Q9. Create a scatter plot using the points $(1,2),(2,4),(3,6)$ and draw the line of best fit. Write the equation of the line of best fit and use it to make a prediction for the value of y when $x=4$.

Q10. Explain the difference between a positive and negative correlation in the context of a scatter plot. Provide an example of each and describe how the line of best fit would be used to make predictions in each case.

Chef's Tips

- Tip 1: Emphasize the importance of creating a clear and accurate scatter plot
- Tip 2: Remind students to use the equation of the line of best fit to make predictions
- Tip 3: Encourage students to check their work and provide opportunities for feedback